COMPLETE LIST

CHARACTERSTICS OF LIST :

1. REPRESENT AS [ ]

2. IT IS AN ORDER SEQUENCE OF ITEAM

3. IT IS MUTABLE DATA STRUCTURE

4. LIST IS HETROGENOUS DONT NEED TO BE OF THE SAME DATA TYPE

5. SLICING AND INDEXING ARE ALLOWED

In [1]:
```
L1 = []        #empty list method 1
```

In [2]:
```
L1
```

Out[2]:
```
[]
```

In [3]:
```
print(type(L1))
```

```
<class 'list'>
```

In [4]:
```
L2 = list()      #empty list method 2
L2
```

Out[4]:
```
[]
```

In [5]:
```
L3 = [ 10,20, True, "new" ]   #list is heterogenous
L3
```

Out[5]:
```
[10, 20, True, 'new']
```

In [6]:
```
len(L3)
```

Out[6]:
```
4
```

In [7]:
```
nest = [10,[20,25],30]  #nested list: within list,one of iteam is list
nest
```

Out[7]: [10, [20, 25], 30]

In [8]:
```
nest[1]
```

Out[8]: [20, 25]

In [9]:
```
nest[1][0]
```

Out[9]: 20

In [10]:
```
L4 = [10,20,30]   #list of integers number
L4
```

Out[10]: [10, 20, 30]

In [11]:
```
L5=[10.75,15.25,60.89]  #list of float number
L5
```

Out[11]: [10.75, 15.25, 60.89]

In [12]:
```
L6=['one','two','three']   #list of string
L6
```

Out[12]: ['one', 'two', 'three']

In [13]:
```
L7=[10,'python',10.55,[5,8],{1:'one',2:'two'}]   #list of mix data type(heteroge
L7
```

Out[13]: [10, 'python', 10.55, [5, 8], {1: 'one', 2: 'two'}]

In [14]:
```
L7[4][1]
```

Out[14]: 'one'

In [15]:
```
L7[3][1]
```

Out[15]: 8

LIST INDEXING AND SLICING

In [16]:
```
L8=[10,20,'ROSHANI','SUNIL',[50,20],{'PYTHPN WITH DATA SCIENCE'}]
L8
```

Out[16]: [10, 20, 'ROSHANI', 'SUNIL', [50, 20], {'PYTHPN WITH DATA SCIENCE'}]

In [17]:
```
L8[0] #RETREIVE FIRST ELEMENT OF THE LIST
```

Out[17]: 10

In [18]:
```
L8[-1]  #RETREIVE LAST ITEAM OF THE LIST
```

Out[18]: {'PYTHPN WITH DATA SCIENCE'}

In [19]:
```
L8[4][1]  #NESTED INDEXING
```

Out[19]: 20

In [20]:
```
L8
```

Out[20]: [10, 20, 'ROSHANI', 'SUNIL', [50, 20], {'PYTHPN WITH DATA SCIENCE'}]

LIST SLICING:

1. [:] > RETURN ENTIRE LIST

3. [2:5] > START 2nd TO 4th (n-1) INDEX

5. [3:] > START 3rd TO LAST INDEX

7. [:4] > RETURN FIRST 3 ITEAMS

9. [-3:] > RETURN LAST 3 ITEAMS

10. [: : :] > [STRT : END(N-1): STEP]

11. [: :-1] > REVERSE LIST

In [21]:
```
L9=[10,20,30,40,50,60,70]
L9
```

Out[21]: [10, 20, 30, 40, 50, 60, 70]

In [22]:
```
L9[:-1]
```

Out[22]: [10, 20, 30, 40, 50, 60]

In [23]:
```
L9[::-1]
```

Out[23]: [70, 60, 50, 40, 30, 20, 10]

In [24]:
```
L9[0:7:1]
```

Out[24]: [10, 20, 30, 40, 50, 60, 70]

In [25]:
```
L9
```

Out[25]: [10, 20, 30, 40, 50, 60, 70]

LOOP THROUGH A LIST

In [26]:
```
for i in L9 :
   print(i)
```

```
10
20
30
40
50
60
70
```

In [27]:
```
for i in enumerate(L9):
    print(i)
```

```
(0, 10)
(1, 20)
(2, 30)
(3, 40)
(4, 50)
(5, 60)
(6, 70)
```

In [28]:
```
L10 = [10,20,20,'python', 'python',101,200,200,300]
L10
```

Out[28]:
```
[10, 20, 20, 'python', 'python', 101, 200, 200, 300]
```

In [29]:
```
L10.count(200)
```

Out[29]:
```
2
```

In [30]:
```
L10.count('python')
```

Out[30]:
```
2
```

In [31]:
```
L11 = [5,7,2,11,9,4,1,3]
L11
```

Out[31]:
```
[5, 7, 2, 11, 9, 4, 1, 3]
```

In [32]:
```
L11.sort() #BYDEFOULT ASCENDING ORDER
L11
```

Out[32]:
```
[1, 2, 3, 4, 5, 7, 9, 11]
```

In [33]:
```
L11.sort(reverse = True) # decending order
L11
```

Out[33]:
```
[11, 9, 7, 5, 4, 3, 2, 1]
```

In [34]:
```
L11
```

Out[34]:
```
[11, 9, 7, 5, 4, 3, 2, 1]
```

In [35]:
```
L11.remove(11) #REMOVE ITEAM ONLY ONE BY ASSING VALU ONLY
L11
```

Out[35]:
```
[9, 7, 5, 4, 3, 2, 1]
```

In [36]:
```
L11
```

Out[36]:
```
[9, 7, 5, 4, 3, 2, 1]
```

In [37]:
```
L11.pop(0) #REMOVE ITEAM BY INDEX
L11
```

Out[37]:
```
[7, 5, 4, 3, 2, 1]
```

In [38]:
```
L11.pop()   #REMOVE ITEAM AT LAST BYDEFAULT
L11
```

Out[38]:
```
[7, 5, 4, 3, 2]
```

In [39]:
```
L11
```

Out[39]:
```
[7, 5, 4, 3, 2]
```

In [40]:
```
del L11[1]  #remove/ del iteam through index location
L11
```

Out[40]:
```
[7, 4, 3, 2]
```

In [41]:
```
del L11[0]
L11
```

Out[41]:
```
[4, 3, 2]
```

In [42]:
```
L12 = ['ONE','TWO','THREE','FOUR','FIVE','SIX','SEVEN']
L12
```

Out[42]:
```
['ONE', 'TWO', 'THREE', 'FOUR', 'FIVE', 'SIX', 'SEVEN']
```

In [43]:
```
L12.append( 'eight')  #Add value to the end of the list.
L12
```

Out[43]:
```
['ONE', 'TWO', 'THREE', 'FOUR', 'FIVE', 'SIX', 'SEVEN', 'eight']
```

In [44]:
```
L12
```

Out[44]:
```
['ONE', 'TWO', 'THREE', 'FOUR', 'FIVE', 'SIX', 'SEVEN', 'eight']
```

In [45]:
```
L12.insert(3,3) # insert(index,value) ,Insert value before index
L12
```

Out[45]:
```
['ONE', 'TWO', 'THREE', 3, 'FOUR', 'FIVE', 'SIX', 'SEVEN', 'eight']
```

In [46]:
```
L12
```

Out[46]: ['ONE', 'TWO', 'THREE', 3, 'FOUR', 'FIVE', 'SIX', 'SEVEN', 'eight']

In [47]:
```
L12[0]=1
L12[8]=8     #change value of string
L12[4]=4
L12
```

Out[47]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [48]:
```
L12.copy()
L12
```

Out[48]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [49]:
```
L13= L12.copy()
L13
```

Out[49]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [50]:
```
L12=L13
L13
```

Out[50]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [51]:
```
id(L12),id(L13)
```

Out[51]: (2297111887040, 2297111887040)

In [52]:
```
L13
```

Out[52]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [53]:
```
L13.index('SEVEN',0,8)      #to find index of given valu ,index(iteam,start,end)
                                  # element → the value you want to find
                                   # start (optional) → where to start searchin
                                   # end (optional) → where to stop searching
print(L13.index('SEVEN'))
```

7

In [54]:
```
L13.index('SEVEN')          #without mention start,end
print(L13.index ('SEVEN'))
```

7

In [55]:
```
L13
```

Out[55]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8]

In [56]:
```
L4
```

Out[56]: [10, 20, 30]

In [57]:
```
L13.extend(L4)    #add two list
L13
```

Out[57]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8, 10, 20, 30]

In [58]:
```
L14 = L13+L4  #add two list through "+" operator
L14
```

Out[58]:
```
[1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8, 10, 20, 30, 10, 20, 30]
```

In [59]:
```
L14
```

Out[59]:
```
[1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8, 10, 20, 30, 10, 20, 30]
```

LIST MEMBERSHIP

In [60]:
```
'THREE' in L14
```

Out[60]:
```
True
```

In [61]:
```
100 in L14
```

Out[61]:
```
False
```

In [62]:
```
L14
```

Out[62]:
```
[1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8, 10, 20, 30, 10, 20, 30]
```

In [63]:
```
all(L14)  # will return true if one value in list is true(value non zero)
```

Out[63]:
```
True
```

In [64]:
```
new_list =[1,2,3,0]  # will return false if one value in list is false(value : z
new_list
```

Out[64]:
```
[1, 2, 3, 0]
```

All/Any

all() : it return

1.True = if all elements in a list are true

2.false = if all elements in a list are false

any() :return true if any element in the list is true.if not ,any() return false

In [65]:
```
all(new_list)
```

Out[65]: False

In [66]:
```
L14
```

Out[66]: [1, 'TWO', 'THREE', 3, 4, 'FIVE', 'SIX', 'SEVEN', 8, 10, 20, 30, 10, 20, 30]

In [67]:
```
any(L14)
```

Out[67]: True

In [68]:
```
new_list
```

Out[68]: [1, 2, 3, 0]

In [69]:
```
any(new_list)
```

Out[69]: True

In [70]:
```
list16 = [0,0,0,0]
list16
```

Out[70]: [0, 0, 0, 0]

In [71]:
```
any(list16)
```

Out[71]: False

In [ ]:

In [ ]:

In [ ]: